LogiFlow Road Delay Prediction ML

1. Road Pipeline Overview

LogiFlow road optimization combines route planning, traffic analysis, demand forecasting, asset utilization monitoring, delay prediction, and risk-aware route ranking. The objective is to identify shipments likely to experience delays before route execution.

2. Training Data

Dataset: smart_logistics_dataset.csv. Training size: 1,000 logistics events. Target variable: Logistics_Delay (binary classification indicating whether a shipment is likely to be delayed).

3. Model & Validation

Algorithm: HistGradientBoostingClassifier. Features: Traffic Severity, Temperature, Humidity, Asset Utilization, Demand Forecast. Validation: 80/20 stratified holdout split and 5-fold cross validation.

4. Quantifier 1 – Delay Detection Accuracy

Value: 73.0%. Percentage of shipments correctly classified as delayed or on-time. Measured on a stratified holdout dataset.

5. Quantifier 2 – High-Risk Shipment Detection

Value: 68.0%. Percentage of delayed shipments successfully identified before execution. Corresponds to recall on delayed shipments.

6. Quantifier 3 – Cross-Validated Reliability

Value: 78.3%. Average ROC-AUC across five validation folds, representing predictive reliability across independent splits.

7. Error & Evaluation Metrics

Accuracy: 73.0%, Precision: 81.0%, Recall: 68.0%, F1 Score: 74.0%, ROC-AUC: 78.2%, CV ROC-AUC: 78.3%. Accuracy measures overall correctness. Precision measures confidence in predicted delays. Recall measures the ability to identify delayed shipments. F1 balances precision and recall. ROC-AUC measures ranking quality across risk thresholds.

8. Retraining

Retrain the model using the logistics training pipeline, regenerate the model artifact, refresh evaluation metrics, and redeploy the backend endpoint serving Road ML quantifiers.